Exercises Week 2

In this set of self-study exercise we will get familiar with the following concepts:

- Omitted variable bias: the simple case
- Inclusion of irrelevant variables in a regression model
- Multicollinearity
- Dummy variables

1. **Omitted Variable Bias: The Simple Case**: Suppose that $\ln(wage_i)$ is determined by

$$\ln(wage_i) = \beta_1 + \beta_2 educ_i + \beta_3 abil_i + \varepsilon_i \tag{1}$$

where $E(\varepsilon_i | educ_i, abil_i) = 0$. In our dataset $abil$ (ability) is not observed. So, instead we estimate the following model:

$$\ln(wage_i) = \beta_1 + \beta_2 educ_i + \nu_i \tag{2}$$

where $\nu_i = \beta_3 abil_i + \varepsilon_i$

(a) Is $E(\nu_i | educ_i) = 0$ in model 2? Explain.

Solution: Notice that

$$E(\nu_i | educ_i) = E(\beta_3 abil_i + \varepsilon_i | educ_i) = \beta_3 E(abil_i | educ_i) \neq 0$$

because $abil_i$ and $educ_i$ are positively correlated ($E(abil_i | educ_i) > 0$). People with more years of schooling are likely to have higher ability.

(b) Given your answer in the previous part, is assumption 1.2 satisfied in equation (2)?

Solution: In case of equation (2), assumption 1.2 (strict exogeneity) is violated! In other words, a problem of **omitted variable bias** arises.

(c) Let

$\tilde{\beta}_2$ be the estimator of β_2 from a simple regression of $\ln(wage_i)$ on $educ_i$ (cf. equation (2)).
$y_i = \ln(wage_i)$, $x_{2i} = educ_i$ and $x_{3i} = abil_i$

Model (1) can be rewritten as follows:

$$y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i \tag{3}$$

(equation (2) can be rewritten in an analogous way.) Then it can be shown that

$$E(\tilde{\beta}_2 | x_{2i}, x_{3i}) = \beta_2 + \beta_3 \tilde{\delta}_1 \tag{4}$$

where $\tilde{\delta_1}$ is the estimate for δ_1 from the following regression

$$x_{3i} = \delta_0 + \delta_1 x_{2i} + \zeta_i \tag{5}$$

The proof of this statement will be given later in the course. Equation (4) implies that the **omitted variable bias** resulting from omitting $\textit{ability}_i$ from the model is equal to

$$bias(\tilde{\beta}_2 | x_2, x_3) = E(\tilde{\beta}_2 | x_2, x_3) - \beta_2 = \beta_3 \tilde{\delta}_1 \tag{6}$$

Given equation (6), what are the two scenarios under which **no bias** occurs (i.e. we have an unbiased estimator)? How can these scenarios be interpreted in the wage example?

Solution: From equation (6), we see that there are two cases where the estimator $\tilde{\beta}_2$ is unbiased.

- $\beta_3 = 0$
- $\tilde{\delta}_1 = 0$

In the wage example, the two conditions imply that ability has no impact on the hourly wage rate or equivalently the variables x_2 (education) is uncorrelated with the omitted variable x_3 (ability).

(d) What do you think the signs of β_3 and δ_1 are in the wage example? Based on the expression for the bias (equation (6)), do you expect that the estimator $\tilde{\beta}_2$ is **biased upwards** (positive bias) or **biased downwards** (negative bias)?

Solution: In the wage example above, one should expect that

- $\beta_3 > 0$
- $\delta_1 > 0$ (positive correlation between years of schooling and ability)

Note that in this case, the estimator $\tilde{\beta}_2$ is **biased upwards** because the bias is positive. Also note that in this example:

$$E(\nu_i | educ_i) = E(\beta_3 abil_i + \varepsilon_i | educ_i) = \beta_3 E(abil_i | educ_i) > 0$$

(e) Using the data set wage1.dta estimate the model in equation (2) in Stata. What is the estimated effect of years of schooling on hourly wage?

Solution:

```
. gen lwage=ln(wage)

. reg lwage educ

      Source |       SS           df       MS      Number of obs   =       526
-------------+----------------------------------   F(1, 524)       =    119.58
       Model |  27.5606288         1  27.5606288   Prob > F        =    0.0000
    Residual |  120.769123       524  .230475425   R-squared       =    0.1858
-------------+----------------------------------   Adj R-squared   =    0.1843
       Total |  148.329751       525   .28253286   Root MSE        =    .48008

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0827444   .0075667    10.94   0.000     .0678796    .0976091
       _cons |   .5837727   .0973358     6.00   0.000     .3925563    .7749891
------------------------------------------------------------------------------
```

0.083: A one year increase in schooling, increases hourly wage by 8.3 percent.

(f) In part d you should have found that the bias is positive and thus the estimator $\tilde{\beta}_2$ is **biased upwards**. Does this mean that the true return to schooling is lower than the estimated β_2?

Solution: We estimated the returns to schooling to be equal to 0.083. This estimation result from only a single sample, so we cannot say that .083 is greater than β_2. The true return to education could be lower or higher than 8.3 percent (and we will never know for sure). Nevertheless, **we know that the average of the estimates across all random samples would be too large**.

2. **Including Irrelevant Variables in a Regression Model** Suppose we specify the model as

$$y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + \varepsilon_i \tag{7}$$

and this model satisfies Assumptions 1.1 through 1.4. However, $\beta_4 = 0$. In other words, the "true" model is

$$y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i \tag{8}$$

Let

$\mathbf{x}_j = (x_{1j}, \ldots, x_{nj})'$, $j = 2, ..., 4$: $(n \times 1)$ vectors
$\tilde{\mathbf{X}} = (\boldsymbol{\iota}_n, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4)$
$\mathbf{X} = (\boldsymbol{\iota}_n, \mathbf{x}_2, \mathbf{x}_3)$
$\hat{\boldsymbol{\beta}} = (\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\mathbf{y}$
$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$

So, $\hat{\boldsymbol{\beta}}$ be the OLS estimator of regression (7) and $\mathbf{b}$ the OLS estimator stemming from the 'true' model (8).

(a) Does inclusion of x_{4i} affects unbiasdness of $\hat{\boldsymbol{\beta}}$, so the OLS estimator of (7)? Hint: think in terms of the assumptions needed to prove unbiasdness of b

Solution: No, none of the assumptions are violated. One can show that

$$E(\hat{\boldsymbol{\beta}}|\tilde{\mathbf{X}}) = E((\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\mathbf{y}|\tilde{\mathbf{X}}) = \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ 0 \end{pmatrix}$$

(In the last step, we use $\beta_4 = 0$). Consequently, by the law of total expectations

$$E(\hat{\boldsymbol{\beta}}) = E(E(\hat{\boldsymbol{\beta}}|\mathbf{X})) = E(\boldsymbol{\beta}) = \boldsymbol{\beta}$$

In other words, $\hat{\boldsymbol{\beta}}$ is an unbiased estimator.

(b) Does inclusion of x_{4i} affects efficiency of $\hat{\boldsymbol{\beta}}$, so the OLS estimator of (7)? Hint: use Gauss Markov Theorem

Solution: **b** is unbiased (proposition 1.1a) and linear (in **y**). Then, from Gauss Markov Theorem we know that **b** is BLUE. Therefore, **b** is more efficient for $(\beta_1, \beta_2, \beta_3)'$ than $\hat{\beta}$.
More intuitive explanation: Recall **Proposition 1.1.b**: Under assumptions 1.1-1.4,

$$Var(\mathbf{b}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \tag{9}$$

Using the Frisch-Waugh theorem (see Hayashi for details) we can show that the elements on the main diagonal of $Var(\mathbf{b}|\mathbf{X})$ can be written as

$$Var(b_j|\mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_{ji} - \bar{x}_j)^2(1 - R_j^2)}, \; j = 2, \ldots, K \tag{10}$$

where R_j^2 is the R-squared from regressing x_j on all other independent variables (and including an intercept).
From equations (9) and (10), it becomes clear that the conditional variance of b_j depends on three factors

1. σ^2 the variance of the error term
   $\sigma^2 \uparrow \Rightarrow Var(b_j|\mathbf{X}) \uparrow$

2. The total sample variation $SST_j = \sum_{i=1}^{n}(x_{ji} - \bar{x}_j)^2$
   $SST_j \uparrow \Rightarrow Var(b_j|\mathbf{X}) \downarrow$
   (Notice that SST_j increases with sample size!)

3. R_j^2: the linear relationships among the independent variables
   $R_j^2 \uparrow \Rightarrow Var(b_j|\mathbf{X}) \uparrow$

If we have many regressors that are highly correlated (high R_j) then we have a problem of **multicollinearity**.

Notice a distinction with the book:
Contrary to Hayashi, I call high (but not perfect) correlation between two or more independent variables **multicollinearity** (high R_j^2 for at least one of the explanatory variables).
Perfect multicollinearity refers to the case that for at least one j $R_j^2 = 1$. In that case assumption 1.3 (data matrix X is of full column rank) is violated (Hayashi calls this multicollinearity)

(c) Why do we care about efficiency?

Solution: The size of $Var(b_j|\mathbf{X})$ is practically important. A larger variance means a less precise estimator, and this translates into larger confidence intervals and less accurate hypotheses tests (next week).

3. **Multicollinearity**. Now consider the following model:

$$\ln(wage_i) = \beta_1 + \beta_2 educ_i + \beta_3 exper_i + \beta_4 exper_i^2 + \varepsilon_i \tag{11}$$

where $exper_i$ denotes the years of experience in the labor market.

(a) Write down the expression for the returns to experience.

Solution: In this specification the returns to experience

$$\frac{\partial \ln(wage_i)}{\partial exper_i} = \beta_3 + 2\beta_4 exper_i$$

(b) Using Stata and the data set *wage1.dta*, estimate the model in equation (11).

Solution:

```
. gen exper2=exper^2

. reg lwage educ exper exper2

      Source |       SS           df       MS      Number of obs   =       526
-------------+----------------------------------   F(3, 522)       =     74.67
       Model |  44.5393713         3  14.8464571   Prob > F        =    0.0000
    Residual |   103.79038       522  .198832146   R-squared       =    0.3003
-------------+----------------------------------   Adj R-squared   =    0.2963
```

```
       Total |  148.329751       525   .28253286   Root MSE        =    .44591

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0903658    .007468    12.10   0.000     .0756948    .1050368
       exper |   .0410089   .0051965     7.89   0.000     .0308002    .0512175
      exper2 |  -.0007136   .0001158    -6.16   0.000     -.000941   -.0004861
       _cons |   .1279975   .1059323     1.21   0.227    -.0801085    .3361035
------------------------------------------------------------------------------
```

(c) Are coefficients β_3 and β_4 precisely estimated (i.e. do they have a small variance)? Why do you think this is the case, i.e. which feature of the data is responsible for the precise (or imprecise) estimates? Consider the formula for the variance of **b** to answer this question.

Solution: Yes, β_3 and β_4 are precisely estimated. This is due to large variation in $exper$ as demonstrated in the table below:

Solution:

```
. su exper

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
       exper |        526    17.01711    13.57216          1         51
```

(d) Are the estimated returns to experience positive or negative?

Solution: $\frac{\partial log(wage_i)}{\partial exper_i} > 0$ if $exper < 29 (\approx -.0410089/(2 * -.0007136)$ The returns are positive upto 29 years of experience.

(e) Now consider an alternative dataset which only covers young employees. Before looking at the data, do you expect that the coefficients β_3 and β_4 will be more or less precisely estimated in this sample?

Solution: Less, as by construction the variance of exper will be lower in this sample as we only have young workers.

(f) Using data set *wage2.dta*, estimate model in equation (11) and compare the precision of the estimates of the coefficients β_3 and β_4 with the estimates from part b.

Solution:

```
 gen exper2=exper^2

. reg lwage educ exper exper2
```

```
      Source |       SS           df       MS      Number of obs   =       935
-------------+----------------------------------   F(3, 931)       =     46.75
       Model |   21.688779         3  7.22959299   Prob > F        =    0.0000
    Residual |  143.967504       931  .154637491   R-squared       =    0.1309
-------------+----------------------------------   Adj R-squared   =    0.1281
       Total |  165.656283       934  .177362188   Root MSE        =    .39324

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0779866   .0066242    11.77   0.000     .0649864    .0909867
       exper |    .016256     .01354     1.20   0.230    -.0103164    .0428285
      exper2 |    .000152    .000567     0.27   0.789    -.0009607    .0012647
       _cons |   5.517432   .1248186    44.20   0.000     5.272474    5.762391
------------------------------------------------------------------------------
```

The estimates are imprecise in this sample. Variance is much higher. This is due to low variation of exper variable as demonstrated below:

```
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
       exper |        935    11.56364    4.374586          1         23
```

(g) Regress $exper^2$ on all rhs variables and consider the R^2 from this regression. What does this indicate?

Solution:

```
. reg exper2 exper educ

      Source |       SS           df       MS      Number of obs   =       935
-------------+----------------------------------   F(2, 932)       =   9541.56
       Model |  9849570.27         2  4924785.14   Prob > F        =    0.0000
    Residual |  481043.032       932  516.140593   R-squared       =    0.9534
-------------+----------------------------------   Adj R-squared   =    0.9533
       Total |  10330613.3       934  11060.6138   Root MSE        =    22.719

------------------------------------------------------------------------------
      exper2 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       exper |   23.15904   .1908916   121.32   0.000     22.78441    23.53366
        educ |  -1.345747   .3801563    -3.54   0.000    -2.091808   -.5996854
       _cons |  -96.84333   6.475965   -14.95   0.000    -109.5525   -84.13417
------------------------------------------------------------------------------
```

The R^2 is 0.95 which means that 95% of variation of $exper^2$ is explained with the other rhs variables. This means that we face a **multicollinearity** problem. Consequently, the parameters β_3 and β_4 aren't estimated with great precision. Given the limited variability, it might be wise to drop the variable $exper^2$ from specification (11)

(h) Estimate the model again but without variable $exper^2$. Did anything change with respect to precision of β_3?

Solution:

```
. reg lwage educ exper

      Source |       SS           df       MS      Number of obs   =       935
-------------+----------------------------------   F(2, 932)       =     70.16
       Model |  21.6776613         2  10.8388306   Prob > F        =    0.0000
    Residual |  143.978622       932    .1544835   R-squared       =    0.1309
-------------+----------------------------------   Adj R-squared   =    0.1290
       Total |  165.656283       934  .177362188   Root MSE        =    .39304

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |    .077782   .0065769    11.83   0.000     .0648748    .0906892
       exper |   .0197768   .0033025     5.99   0.000     .0132956     .026258
       _cons |    5.50271    .112037    49.12   0.000     5.282836    5.722584
------------------------------------------------------------------------------
```

Now, the coefficient β_3 is estimated with precision.

4. **Dummy variables** Consider again the original data set consisting of 526 observations (*wage1.dta*). Extend the model with one dummy independent variable $nonwhite$. The dummy variable $nonwhite$ is defined as follows:

$$\begin{aligned} nonwhite_i &= 1 \text{ if respondent } i \text{ is non-white} \\ &= 0 \text{ otherwise} \end{aligned}$$

(a) Estimate the extended model in Stata.

Solution:

```
. /* Dummy variables */
. gen log_wage=ln(wage)
. gen expersq=exper^2
. reg log_wage  educ exper expersq nonwhite
      Source |       SS       df       MS              Number of obs =     526
-------------+------------------------------           F(  4,   521) =   55.90
       Model |   44.545379     4  11.1363448           Prob > F      =  0.0000
    Residual |  103.784372   521   .19920225           R-squared     =  0.3003
-------------+------------------------------           Adj R-squared =  0.2949
       Total |  148.329751   525   .28253286           Root MSE      =  .44632
------------------------------------------------------------------------------
    log_wage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |    .090251   .0075041    12.03   0.000      .075509    .1049931
       exper |   .0410322   .0052031     7.89   0.000     .0308106    .0512538
     expersq |  -.0007142   .0001159    -6.16   0.000    -.0009419   -.0004864
    nonwhite |  -.0111807    .064382    -0.17   0.862    -.1376609    .1152995
       _cons |   .1304809   .1069908     1.22   0.223    -.0797054    .3406671
------------------------------------------------------------------------------
```

(b) Interpret the coefficient on $educ$.

Solution: One extra year of schooling increases the wage by 9% *keeping other factors constant (ceteris paribus)*. Note that we have a log-level specification.

(c) Interpret the coefficient on $nonwhite$.

Solution: Interpretation $b_{nonwhite}$: a nonwhite person earns about 1.1 percent (= -.0111807*100) less than a white person (reference group) *keeping other factors constant (ceteris paribus)*.

(d) Instead of the dummy variable $nonwhite$, include the dummy variable $female$ in order to analyze the gender wage gap and estimate the model again. What is the estimated wage gap?

Solution:

```
. reg log_wage  educ exper expersq female

      Source |       SS       df       MS              Number of obs =     526
-------------+------------------------------           F(  4,   521) =   86.69
       Model |  59.2711314     4  14.8177829           Prob > F      =  0.0000
    Residual |    89.05862   521   .17093785           R-squared     =  0.3996
-------------+------------------------------           Adj R-squared =  0.3950
       Total |  148.329751   525   .28253286           Root MSE      =  .41345

------------------------------------------------------------------------------
    log_wage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0841361   .0069568    12.09   0.000     .0704692    .0978029
       exper |     .03891   .0048235     8.07   0.000      .029434    .0483859
     expersq |   -.000686   .0001074    -6.39   0.000     -.000897   -.0004751
      female |  -.3371868   .0363214    -9.28   0.000    -.4085411   -.2658324
       _cons |    .390483   .1022096     3.82   0.000     .1896894    .5912767
------------------------------------------------------------------------------
```

Interpretation b_{female}: a female employee earns

$$(e^{-.3371868} - 1) * 100 = -28.622451 \text{ percent}$$

more (i.e. 28.622451 percent less) than a male employee ceteris paribus.

(e) Now add the dummy variable $married$ to the regression equation presented above (next to the dummy $female$). Interpret the results on the $married$ dummy.

Solution:

```
. reg log_wage  educ exper expersq female married
      Source |       SS       df       MS              Number of obs =     526
-------------+------------------------------           F(  5,   520) =   70.00
       Model |  59.6726652     5   11.934533           Prob > F      =  0.0000
    Residual |  88.6570862   520  .170494397           R-squared     =  0.4023
-------------+------------------------------           Adj R-squared =  0.3966
       Total |  148.329751   525   .28253286           Root MSE      =  .41291
------------------------------------------------------------------------------
```

```
    log_wage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0828163   .0070008    11.83   0.000     .0690629    .0965696
       exper |   .0357493   .0052391     6.82   0.000      .025457    .0460417
     expersq |   -.000632   .0001129    -5.60   0.000    -.0008537   -.0004103
      female |  -.3290458   .0366601    -8.98   0.000    -.4010658   -.2570257
     married |   .0645607    .042069     1.53   0.125    -.0180854    .1472068
       _cons |   .3921025   .1020824     3.84   0.000     .1915579    .5926472
------------------------------------------------------------------------------
```

Married individuals make about 6.5% more per hour than unmarried workers, ceteris paribus. In other words, the marriage premium is estimated to be about 6.5 %

(f) Empirical evidence suggests that the marriage premium tends to be different for men and women. To capture this feature in our model, we need to interact the two variables, i.e. include a variable *married*female* in our model. Write down the new model and give the expression for the marriage premium in the new model. (Hint: one way to include the interaction term in Stata is the following:
`i.female##i.married`)

Solution: The new model becomes:

$$\ln(wage_i) = \beta_1 + \beta_2 educ_i + \beta_3 exper_i + \beta_4 exper_i^2 + \beta_5 female_i + \beta_6 married_i + \beta_7 female * married_i + \varepsilon_i \tag{12}$$

The marriage premium is now equal to:

$$\frac{\partial wage}{\partial married} = \beta_6 + \beta_7 * female_i$$

The premium now differs between men and women.

(g) Estimate the extended model. What is the estimated marriage premium?

Solution:

```
 reg lwage educ exper exper2 i.female##i.married

      Source |       SS           df       MS      Number of obs   =       526
-------------+----------------------------------   F(6, 519)       =     64.30
       Model |  63.2456419         6  10.5409403   Prob > F        =    0.0000
    Residual |  85.0841096       519  .163938554   R-squared       =    0.4264
-------------+----------------------------------   Adj R-squared   =    0.4198
       Total |  148.329751       525   .28253286   Root MSE        =    .40489

------------------------------------------------------------------------------
         lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
---------------+----------------------------------------------------------------
          educ |   .0822455    .006866    11.98   0.000     .0687569     .095734
         exper |   .0348695   .0051408     6.78   0.000     .0247701    .0449688
        exper2 |  -.0006192   .0001107    -5.59   0.000    -.0008367   -.0004017
      1.female |  -.1207069   .0573047    -2.11   0.036    -.2332847   -.0081292
     1.married |   .2459701   .0566721     4.34   0.000     .1346352     .357305
```

```
                |
female#married  |
          1 1   |  -.3430929   .0734915    -4.67   0.000    -.4874703   -.1987155
                |
         _cons  |   .2841051   .1027389     2.77   0.006       .08227    .4859403
------------------------------------------------------------------------------
```

The marriage premium is equal to $(e^{0.2459701} - 1) * 100 = 28$percent for men while married women make on average 10 percent less than unmarried women, ceteris paribus.

(h) Sometimes we have categorical regressors. For example, in wage equation, we often control for region of residence. In our example, we distinguish 4 different regions ($g = 4$):

1. North-east (reference group)
2. North-central
3. West
4. South

If a variable takes on g categories, then we need to include $g - 1$ dummy variables in the model. In our case, $g = 4$ so we will include 3 dummy variables. First we choose one category to be the **base** category. Assume "North-east" is the reference category. Estimate the model (without the interaction term) with the three dummy variables for the region of residence. Interpret the coefficient on variable $south$.

Solution:

```
. reg log_wage  educ exper expersq female married northcen south west
      Source |       SS       df       MS              Number of obs =     526
-------------+------------------------------           F(  8,   517) =   45.95
       Model |  61.6387993     8  7.70484991           Prob > F      =  0.0000
    Residual |  86.6909521   517  .167680758           R-squared     =  0.4156
-------------+------------------------------           Adj R-squared =  0.4065
       Total |  148.329751   525   .28253286           Root MSE      =  .40949

------------------------------------------------------------------------------
    log_wage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0808207   .0070101    11.53   0.000      .067049    .0945925
       exper |   .0363615   .0052269     6.96   0.000     .0260929    .0466301
     expersq |   -.000645   .0001128    -5.72   0.000    -.0008665   -.0004235
      female |  -.3345661   .0364315    -9.18   0.000     -.406138   -.2629941
     married |   .0711934   .0417725     1.70   0.089    -.0108712     .153258
    northcen |   -.070182   .0519674    -1.35   0.177    -.1722752    .0319111
       south |  -.1162238   .0486825    -2.39   0.017    -.2118637   -.0205839
        west |     .04643   .0576308     0.81   0.421    -.0667894    .1596494
       _cons |   .4625799   .1074952     4.30   0.000     .2513988     .673761
------------------------------------------------------------------------------
```

Interpretation b_{south}: an employee living in the Southern part of the US earns about 11.6 % less than an employee living in the North East (reference group) keeping other factors fixed. Notice that this means that our regression model is to have different intercepts for, g categories, along with the intercept.

(i) Why do we need to skip one category?

Solution: Due to perfect multicollinearity.

5. (to be completed in R) Return to the simulation exercises from the previous week.
   (a) Recall the formula for the OLS estimator $\mathbf{b} = (X'X)^{-1}X'y$. Program the OLS estimator in R (denote it as b). Obtain the OLS estimates for the model parameters (the $\beta's$). Obtain the fitted values $\hat{y}$ and residuals e.
   (b) Using in-build command lm, check if you obtained the same values of b in your sample. Compare the values to the true parameter values.
   (c) Generate a new error term such that $\varepsilon_{1i} = \varepsilon_i + 0.5 * Si$ where ε_i denotes the original error term defined in this simulation exercise. Which assumption is now violated? Compute the OLS estimates again. How does it compare to the previous estimates? How does it compare to the true parameter values? Is this what you expected? What happens if $\varepsilon_{1i} = \varepsilon_i + 0.00000005 * Si$?
   (d) Now, recall the formula for the estimator of the variance of **b**. Program it in R and to check whether you've done it correctly compare the standard errors you obtained with the standard errors obtained via the build in command lm.
   (e) Go back to the simulation of the independent variables in model. Increase the variance of the variable schooling (S_i). How does this affect the estimated **b** and its standard errors? Is this what you expected?
   (f) Now, go back to the simulation of the error term. Make the variance of the error term a function of S_i.Generate a new dependent variable taking this new error term into account. How does this affect the estimated **b** and its standard errors? Is this what you expected? Are the estimates identical (yes/no/why)?
   (g) Estimate each model 10000 times, each time on a different sample. For both models, compute the average of these estimates. What do you observe? Explain.

Solution: The solution also includes the part from week 1 as it is needed to complete this part.

```
##############################################
#week 1 programming practice
##############################################

# Set seed for reproducibility
set.seed(123)

# Number of observations
n <- 500

# Step 3: Generate explanatory variables
Si <- round(runif(n, min = 8, max = 20))  # Years of Schooling (between 8 and 20 years)
tenure <- round(runif(n, min = 0, max = 30))  # Years of job tenure (between 0 and 30 years)
exper <- round(runif(n, min = 0, max = 40))  # Work experience (between 0 and 40 years)
exper_sq <- exper^2  # Square of experience to capture non-linear effects
```

```
#check for perfect multicolinearity
X<-data.frame(1,Si,tenure,exper, exper_sq)
qr_X <- qr(X)
rank_X <- qr_X$rank

# Step 4: Generate the error term
epsilon <- rnorm(n, mean = 0, sd = 1)  # Normally distributed error term with mean 0 and SD = 1.
#Note that the error term does not have to be normally distributed. However, the mean of the error has

# Step 5: Generate the dependent variable (log of wage)
beta1 <- 1      # Intercept
beta2 <- 0.08   # Effect of schooling
beta3 <- 0.03   # Effect of tenure
beta4 <- 0.04   # Effect of experience
beta5 <- -0.001 # Diminishing returns to experience

ln_wage <- beta1 + beta2 * Si + beta3 * tenure + beta4 * exper + beta5 * exper_sq + epsilon

# Create a data frame to store the generated data
data <- data.frame(Si, tenure, exper, exper_sq, ln_wage, epsilon)

# Print the first few rows of the data to see if it makes sense (do you see any strange values?)
head(data)

# Compute the variance-covariance matrix of epsilon
var_cov_epsilon <- cov(matrix(epsilon, ncol = 1))
print(var_cov_epsilon)

#generate an example of an error term that violates assumption 1.4
epsilon_het <- rnorm(n, mean = 0, sd = sqrt(Si))


#####################################################################
# Week 2
#####################################################################
# Create the design matrix X with a column of ones for the intercept
X <- model.matrix(~ Si + tenure + exper + exper_sq, data = data)

# Extract the dependent variable y
y <- data$ln_wage

# Calculate the OLS estimator b
b <- solve(t(X) %*% X) %*% t(X) %*% y

# Print the OLS estimates
print("OLS Estimates:")
print(b)
# Calculate fitted values y_hat
y_hat <- X %*% b
# Calculate residuals e
e <- y - y_hat

# Fit the model using lm (build in regression function)
model_lm <- lm(ln_wage ~ Si + tenure + exper + exper_sq, data = data)
# Print the summary of the lm model
summary(model_lm)
```

```
# Compare with true parameter values
true_beta <- c(1, 0.08, 0.03, 0.04, -0.001)
print("True Beta Values:")
print(true_beta)

# Generate new error term. By making it a function of Si, we violate assumption 1.2
epsilon_1 <- data$epsilon + 0.5 * data$Si

# Generate new dependent variable
ln_wage_1 <- beta1 + beta2 * Si + beta3 * tenure + beta4 * exper + beta5 * exper_sq + epsilon_1
# Calculate the OLS estimator b (check with the lm function)
b1 <- solve(t(X) %*% X) %*% t(X) %*% ln_wage_1

# Repeat with a very small coefficient
epsilon_2 <- data$epsilon + 0.00000005 * data$Si
ln_wage_2 <- beta1 + beta2 * Si + beta3 * tenure + beta4 * exper + beta5 * exper_sq + epsilon_2
b2 <- solve(t(X) %*% X) %*% t(X) %*% ln_wage_2


#Return ot the original model. Estimate sigma squared (variance of the error term)
#Estimate the original model again
K<-ncol(X)
s2<-as.numeric(t(e)%*%e/(n-K))
# Calculate the variance-covariance matrix of b
var_b <- s2*solve(t(X) %*% X)
# Extract standard errors
se_b <- sqrt(diag(var_b))

# Print standard errors
print("Standard Errors of OLS Estimates:")
print(se_b)

# Compare with lm standard errors
print("Standard Errors from lm:")
print(summary(model_lm)$coefficients[, 2])

# Increase the variance of Si
Si_high <- round(runif(n, min = 6, max = 26))
X_high <- model.matrix(~ Si_high + tenure + exper + exper_sq, data = data)
# Calculate the OLS estimator b
b_high <- solve(t(X_high) %*% X_high) %*% t(X_high) %*% y

# For comparison, decrease the variance of Si (this should increase the standard errors)
Si_low <- round(runif(n, min = 12, max = 16))
X_low <- model.matrix(~ Si_low + tenure + exper + exper_sq, data = data)
# Calculate the OLS estimator b
b_low <- solve(t(X_low) %*% X_low) %*% t(X_low) %*% y

# Generate heteroskedastic error term
epsilon_het <- rnorm(n, mean = 0, sd = sqrt(data$Si))
# Generate new dependent variable with heteroskedasticity
X <- model.matrix(~ Si + tenure + exper + exper_sq, data = data)
ln_wage_het <- beta1 + beta2 * data$Si + beta3 * data$tenure + beta4 * data$exper + beta5 * data$exper_
b_het <- solve(t(X) %*% X) %*% t(X) %*% ln_wage_het

#compare b and b_het
```

```
c<-cbind(b,b_het)
c

# Generate age variable
age <- data$Si + data$exper + 4
# Add age to the data frame
X1 <- model.matrix(~ Si + tenure + exper + exper_sq +age, data = data)
b1 <- solve(t(X1) %*% X1) %*% t(X1) %*% y
#We cannot compute b since assumption 1.3 is now violated

################### Loop ##############################
# Number of observations
n <- 500

# True parameters
beta1 <- 1      # Intercept
beta2 <- 0.08   # Effect of schooling
beta3 <- 0.03   # Effect of tenure
beta4 <- 0.04   # Effect of experience
beta5 <- -0.001 # Diminishing returns to experience

# Number of simulations
n_sims <- 10000

# Arrays to store results
betas_homo <- matrix(NA, nrow = n_sims, ncol = 5)
betas_hetero <- matrix(NA, nrow = n_sims, ncol = 5)

# Loop over simulations
for (i in 1:n_sims) {
  # Generate new explanatory variables
  Si <- round(runif(n, min = 8, max = 20))  # Years of schooling
  tenure <- round(runif(n, min = 0, max = 30))  # Years of job tenure
  exper <- round(runif(n, min = 0, max = 40))  # Work experience
  exper_sq <- exper^2  # Square of experience

  # Construct the design matrix X
  X <- model.matrix(~ Si + tenure + exper + exper_sq)

  # True relationship: y_true = X
  y_true <- X %*% c(beta1, beta2, beta3, beta4, beta5)

  # Generate homoskedastic error term
  epsilon_homo <- rnorm(n, mean = 0, sd = 1)

  # Generate heteroskedastic error term
  epsilon_hetero <- rnorm(n, mean = 0, sd = sqrt(Si))

  # Generate dependent variables
  y_homo <- y_true + epsilon_homo
  y_hetero <- y_true + epsilon_hetero

  #Compute the OLS estimates
  b_homo <- solve(t(X) %*% X) %*% t(X) %*% y_homo
  b_hetero <- solve(t(X) %*% X) %*% t(X) %*% y_hetero
```

```
  # Store coefficients
  betas_homo[i, ] <- b_homo
  betas_hetero[i, ] <- b_hetero

}

# Calculate the mean of the coefficients across simulations
mean_betas_homo <- colMeans(betas_homo)
mean_betas_hetero <- colMeans(betas_hetero)

# Print the results
cat("Average OLS Estimates (Homoskedastic):\n")
print(mean_betas_homo)

cat("\nAverage OLS Estimates (Heteroskedastic):\n")
print(mean_betas_hetero)

# True parameters
cat("\nTrue Parameters:\n")
print(c(beta1, beta2, beta3, beta4, beta5))
```